

SOLUTIONS_LEVEL3

SOLUTIONS LEVEL 3

ADVANCED MCQs – JEE MAIN STYLE (~130 QUESTIONS)

SECTION 1: NUMERICAL–CONCEPTUAL BASICS (Q1–Q35)

- 10 g of urea ($M = 60 \text{ g mol}^{-1}$) is dissolved in 90 g of water. Mole fraction of urea is approximately
 - 0.018
 - 0.025
 - 0.05
 - 0.10
- 5.85 g of NaCl ($M = 58.5 \text{ g mol}^{-1}$) is dissolved in water to make 1 L of solution. Molarity is
 - 0.05 M
 - 0.10 M
 - 0.50 M
 - 1.0 M
- 36.5 g of HCl ($M = 36.5 \text{ g mol}^{-1}$) is present in 1 L of aqueous solution. The solution is
 - 0.1 M
 - 1 M
 - 10 M
 - 36.5 M
- A 0.5 molal NaCl solution (assume complete dissociation) has effective molality (for colligative properties)
 - 0.25 m
 - 0.5 m
 - 1.0 m
 - 2.0 m
- The mole fraction of water in 1 molal aqueous solution of glucose is ($M_{\text{water}} = 18 \text{ g mol}^{-1}$)
 - ~0.017
 - ~0.018

- C. ~ 0.982
D. 0.5
6. If 0.5 mol of solute A and 1.5 mol of solvent B are mixed, mole fraction of A is
A. 0.25
B. 0.5
C. 0.75
D. 0.33
7. 20 g of a non-volatile solute ($M = 40 \text{ g mol}^{-1}$) is dissolved in 180 g water. Mole fraction of solute is
A. 0.1
B. 0.05
C. 0.02
D. 0.01
8. A solution is prepared by dissolving 0.2 mol of KCl in enough water to make 500 mL solution.
Molarity is
A. 0.1 M
B. 0.2 M
C. 0.4 M
D. 0.5 M
9. To make 250 mL of 0.2 M NaOH from 1 M NaOH, volume of stock solution needed is
A. 25 mL
B. 50 mL
C. 100 mL
D. 200 mL
10. 2 molal solution of glucose in water contains
A. 2 mol glucose in 2 L water
B. 2 mol glucose in 1 kg water
C. 2 mol glucose in 1 L solution
D. 2 mol glucose in 500 g water
11. A solution is 20% (w/w) NaOH. 100 g of this solution contains
A. 20 g NaOH
B. 20 g water

C. 80 g NaOH

D. 50 g NaOH

12. If density of a solution is 1.2 g mL^{-1} and it is 10% (w/w) solute, then mass of solute in 250 mL of solution is

A. 25 g

B. 30 g

C. 20 g

D. 10 g

13. In a binary liquid solution with 1 mol of A and 3 mol of B, mole fraction of B is

A. 0.25

B. 0.33

C. 0.75

D. 0.5

14. Mole fraction of solvent in a solution is 0.8. Mole fraction of solute is

A. 0.2

B. 0.8

C. 1.2

D. 0.02

15. A gas obeys Henry's law with constant k_H . If its solubility at 1 atm is S, then solubility at 3 atm is

A. $S/3$

B. S

C. 3S

D. 9S

16. At 298 K, solubility of a gas is 0.02 mol L^{-1} under 2 atm. Its Henry's constant k_H (in $\text{atm}\cdot\text{L mol}^{-1}$) is

A. 0.01

B. 0.04

C. 100

D. 50

17. For two gases A and B in same solvent at same T, if $k_H(A) > k_H(B)$, then

A. A is more soluble

B. B is more soluble

- C. Both have same solubility
D. Solubility independent of k_H
18. For an ideal binary solution of volatile liquids A and B, total vapour pressure is
- A. $p = x_A p_A^\circ + x_B p_B^\circ$
B. $p = p_A^\circ + p_B^\circ$
C. $p = x_A p_B^\circ + x_B p_A^\circ$
D. $p = x_A x_B (p_A^\circ + p_B^\circ)$
19. For a solution of non-volatile solute in volatile solvent, relative lowering of vapour pressure $\frac{p^\circ - p}{p^\circ}$ is equal to
- A. Mole fraction of solvent
B. Mole fraction of solute
C. Molality
D. Molarity
20. A solution obeying Raoult's law at all compositions and temperatures is
- A. Ideal solution
B. Non-ideal with positive deviation
C. Non-ideal with negative deviation
D. Azeotropic mixture
21. If $\Delta H_{\text{mix}} = 0$ and $\Delta V_{\text{mix}} = 0$ for a solution, it is
- A. Strongly non-ideal
B. Ideal
C. Azeotropic
D. Immiscible
22. Elevation in boiling point is given by $\Delta T_b = K_b m$. This relation holds well for
- A. Concentrated solutions
B. Dilute solutions
C. Pure liquids
D. Colloids
23. Depression in freezing point is proportional to
- A. Mole fraction of solute
B. Molality of solute

C. Molarity of solute

D. Mass of solute

24. Osmotic pressure of an ideal dilute solution at T is given by

A. $\pi = CRT$

B. $\pi = nRT$

C. $\pi = \frac{RT}{c}$

D. $\pi = kT$

25. Colligative properties depend on

A. Chemical nature of solute

B. Number of particles in solution

C. Nature of solvent only

D. Volume of solution only

26. If the vapour pressure of pure solvent is 400 torr and that of solution is 380 torr, mole fraction of solute (non-volatile) is

A. 0.05

B. 0.10

C. 0.95

D. 0.90

27. A 1 molal aqueous solution of solute A has $\Delta T_f = 1.86$ K. For same solvent, another 1 molal solution of B has $\Delta T_f = 0.93$ K. Which statement is correct?

A. B associates, A does not

B. A associates, B does not

C. B dissociates, A does not

D. Both associate

28. For 1 molal NaCl solution (complete dissociation), elevation in boiling point is (for water, $K_b = 0.52$ K kg mol⁻¹)

A. 0.52 K

B. 1.04 K

C. 2.08 K

D. 0.26 K

29. If osmotic pressure of 0.1 M urea solution at 300 K is π , osmotic pressure of 0.1 M NaCl solution (assume $i \approx 2$) at same T is
- A. $\pi/2$
 - B. π
 - C. 2π
 - D. Zero
30. For solution where solute molecules dimerise, observed molar mass is
- A. Half the normal
 - B. Double the normal
 - C. Same as normal
 - D. Triple the normal
31. A non-ideal solution shows positive deviation from Raoult's law when
- A. $\Delta H_{\text{mix}} < 0$
 - B. $\Delta H_{\text{mix}} > 0$
 - C. $\Delta H_{\text{mix}} = 0$
 - D. Solute-solvent interaction is strongest
32. A non-ideal solution shows negative deviation from Raoult's law when
- A. A-B interactions are stronger than A-A, B-B
 - B. A-B interactions are weaker than A-A, B-B
 - C. No new interactions
 - D. Only hydrogen bonding present in pure components
33. For a binary solution of A and B showing positive deviation from Raoult's law, total vapour pressure curve lies
- A. Above ideal line
 - B. Below ideal line
 - C. Coincides with ideal line
 - D. Vertical line
34. In Henry's law, which pair is correct?
- A. Gas in gas
 - B. Gas in liquid
 - C. Liquid in solid
 - D. Solid in liquid (always)

35. Henry's law constant k_H for O_2 in water is higher than that for CO_2 in water. Therefore, at a given pressure
- A. O_2 is more soluble
 - B. CO_2 is more soluble
 - C. Both have same solubility
 - D. No conclusion

SECTION 2: NUMERICAL RAOULT'S LAW / VAPOUR PRESSURE (Q36–Q60)

36. Vapour pressures of pure benzene and toluene at a temperature are 200 mm Hg and 100 mm Hg respectively. In an ideal solution containing 0.5 mole fraction benzene, total vapour pressure is
- A. 150 mm Hg
 - B. 100 mm Hg
 - C. 200 mm Hg
 - D. 50 mm Hg
37. In Q36, partial vapour pressure of benzene is
- A. 100 mm Hg
 - B. 150 mm Hg
 - C. 200 mm Hg
 - D. 50 mm Hg
38. In Q36, mole fraction of benzene in vapour phase is
- A. 0.33
 - B. 0.5
 - C. 0.67
 - D. 0.75
39. Mole fraction of component A in liquid phase is 0.4. Vapour pressure of pure A and B are 300 mm Hg and 200 mm Hg. Total vapour pressure is
- A. 240 mm Hg
 - B. 260 mm Hg
 - C. 280 mm Hg
 - D. 220 mm Hg

40. In Q39, partial vapour pressure of A is
- A. 80 mm Hg
 - B. 120 mm Hg
 - C. 200 mm Hg
 - D. 300 mm Hg
41. Liquid A and B form an ideal solution. $p_A^\circ = 100 \text{ kPa}$, $p_B^\circ = 50 \text{ kPa}$. If total vapour pressure is 80 kPa, mole fraction of A in liquid is
- A. 0.2
 - B. 0.4
 - C. 0.6
 - D. 0.8
42. For solution in Q41, mole fraction of A in vapour phase is approximately
- A. 0.5
 - B. 0.6
 - C. 0.7
 - D. 0.8
43. In an ideal solution of A and B, at some composition, partial vapour pressures of A and B are equal. If $p_A^\circ = 2p_B^\circ$, then mole fraction of A in liquid is
- A. $1/3$
 - B. $2/3$
 - C. $1/2$
 - D. $1/4$
44. A solution contains 1 mol volatile A and 3 mol volatile B forming ideal solution. If $p_A^\circ = 80 \text{ kPa}$, $p_B^\circ = 40 \text{ kPa}$, total vapour pressure is
- A. 40 kPa
 - B. 50 kPa
 - C. 60 kPa
 - D. 80 kPa
45. In Q44, partial vapour pressure of A is
- A. 20 kPa
 - B. 40 kPa

- C. 60 kPa
- D. 80 kPa

46. A solution of non-volatile solute in water has vapour pressure 19 mm Hg at 25 °C, pure water has 20 mm Hg. Mole fraction of solute is
- A. 0.05
 - B. 0.1
 - C. 0.25
 - D. 0.5
47. A solution of non-volatile solute has vapour pressure 95% of pure solvent. Mole fraction of solute is
- A. 0.05
 - B. 0.95
 - C. 0.1
 - D. 0.9
48. In a solution of 0.2 mol solute and 0.8 mol solvent, relative lowering of vapour pressure is
- A. 0.2
 - B. 0.1
 - C. 0.8
 - D. 0.25
49. The vapour pressure of pure solvent is 100 kPa. When non-volatile solute is added, vapour pressure becomes 90 kPa. Mole fraction of solute is
- A. 0.1
 - B. 0.9
 - C. 0.01
 - D. 0.5
50. A solution obeying Raoult's law has mole fraction of solvent 0.9. Vapour pressure of pure solvent is 50 kPa. Vapour pressure of solution is
- A. 5 kPa
 - B. 45 kPa
 - C. 50 kPa
 - D. 0.9 kPa
51. If total vapour pressure of an ideal binary solution at some composition is between p_A° and p_B° , then
- A. It must be azeotropic

- B. It must be ideal
 - C. It must obey Henry's law
 - D. It must have non-volatile solute
52. For a binary ideal solution, the variation of total vapour pressure with composition (mole fraction) is
- A. Linear
 - B. Parabolic
 - C. Exponential
 - D. Discontinuous
53. A solution of A and B shows a maximum in total vapour pressure vs composition curve. It likely forms
- A. Minimum boiling azeotrope
 - B. Maximum boiling azeotrope
 - C. No azeotrope
 - D. Immiscible liquids
54. A solution showing minimum in total vapour pressure vs composition curve likely forms
- A. Minimum boiling azeotrope
 - B. Maximum boiling azeotrope
 - C. No azeotrope
 - D. Ideal solution
55. Vapour pressure of an ideal solution at a given composition is maximum for
- A. Negative deviation
 - B. Positive deviation
 - C. Ideal behaviour
 - D. Non-volatile solute
56. A solution of ethanol–water as compared to ideal behaviour has
- A. Higher vapour pressure
 - B. Lower vapour pressure
 - C. Same vapour pressure
 - D. Cannot be compared
57. A solution of acetone–chloroform as compared to ideal behaviour has
- A. Higher vapour pressure

- B. Lower vapour pressure
 - C. Same vapour pressure
 - D. Cannot be compared
58. If a binary liquid mixture shows positive deviation from Raoult's law, its boiling point compared to pure components is
- A. Lower than both
 - B. Higher than both
 - C. Equal to average
 - D. Unchanged
59. If a binary liquid mixture shows negative deviation from Raoult's law, its boiling point compared to pure components is
- A. Lower than both
 - B. Higher than both
 - C. Equal to average
 - D. Unchanged
60. At the composition of a minimum boiling azeotrope
- A. Total vapour pressure is minimum
 - B. Total vapour pressure is maximum
 - C. Total vapour pressure equals atmospheric pressure only
 - D. Composition of liquid and vapour are different

SECTION 3: COLLIGATIVE PROPERTIES – ADVANCED MCQs (Q61–Q100)

61. 1 mol of non-volatile solute is dissolved in 1000 g of water. $\Delta T_f = 1.86$ K. If ΔT_f observed is 0.93 K, van't Hoff factor i is
- A. 0.5
 - B. 1
 - C. 2
 - D. 3
62. A solution of an electrolyte AB_2 has van't Hoff factor 2.5. Assuming $AB_2 \rightarrow A^{2+} + 2B^-$, degree of dissociation α is
- A. 0.5

- B. 1.0
C. 0.75
D. 0.25
63. Acetic acid associates in benzene to form dimers. For 1 molal solution, ΔT_f observed is half of expected. Van't Hoff factor is
A. 1
B. 0.5
C. 2
D. 0
64. The osmotic pressure of 0.1 M solution of a protein is 1 atm at 300 K. Approximate molar mass is ($R \approx 0.082 \text{ L atm K}^{-1} \text{ mol}^{-1}$, assume 1 L)
A. 24.6 g mol^{-1}
B. 246 g mol^{-1}
C. 2460 g mol^{-1}
D. $24,600 \text{ g mol}^{-1}$
65. A 0.01 M solution of a non-electrolyte has osmotic pressure 0.246 atm at 300 K. If a 0.01 M solution of electrolyte shows 0.492 atm, van't Hoff factor is
A. 0.5
B. 1
C. 2
D. 4
66. Two solutions A and B have same freezing point depression in water. A contains glucose, B contains sucrose. Then
A. Moles of glucose = moles of sucrose
B. Mole fraction of solute same in both
C. Number of solute particles in each solution is same
D. All
67. A solution of urea ($M = 60 \text{ g mol}^{-1}$) is 0.5 molal. Another solution of glucose ($M = 180 \text{ g mol}^{-1}$) shows same freezing point depression. Approximate molality of glucose solution is
A. 0.5 m
B. 1.0 m

- C. 0.25 m
- D. 0.75 m

68. If 18 g of glucose is present in 1 kg water, molality is

- A. 0.1 m
- B. 0.5 m
- C. 1.0 m
- D. 2.0 m

69. A 0.5 molal solution of a non-electrolyte shows $\Delta T_b = 0.26$ K. K_b of solvent is

- A. $0.26 \text{ K kg mol}^{-1}$
- B. $0.52 \text{ K kg mol}^{-1}$
- C. $1.04 \text{ K kg mol}^{-1}$
- D. $0.13 \text{ K kg mol}^{-1}$

70. A solution contains 1 mol of glucose in 1 kg water. Freezing point is -1.86°C . If 1 mol NaCl in 1 kg water gives -3.72°C , this indicates

- A. NaCl associates
- B. NaCl dissociates
- C. Glucose dissociates
- D. Glucose associates

71. A certain solute has normal molar mass 100 g mol^{-1} . From boiling point elevation, molar mass is found to be 50 g mol^{-1} . This indicates

- A. Association
- B. Dissociation (2 ions)
- C. No abnormality
- D. Polymerization

72. Van't Hoff factor for Na_2SO_4 (assume complete dissociation into 3 ions) is

- A. 1
- B. 2
- C. 3
- D. 4

73. A 1 molal solution of Na_2SO_4 in water ($i=3$) has effective molality for colligative properties

- A. 1 m
- B. 2 m

- C. 3 m
- D. 0.33 m

74. If osmotic pressure of 1 m sucrose at 300 K is 24.6 atm, that of 0.5 m sucrose at same T is
- A. 12.3 atm
 - B. 24.6 atm
 - C. 49.2 atm
 - D. 6.15 atm
75. At same T and for same solvent, solutions with same osmotic pressure have
- A. Same molarity (if non-electrolyte)
 - B. Same molality
 - C. Same vapour pressure
 - D. Same density
76. Depression in freezing point is more convenient to measure than elevation in boiling point because
- A. ΔT_f is larger
 - B. ΔT_b is larger
 - C. Freezing point is more stable
 - D. Both A and C
77. Osmotic pressure measurement is especially useful for
- A. Low molar mass solutes only
 - B. High molar mass solutes (polymers, proteins)
 - C. Gaseous solutes only
 - D. Non-aqueous solutions only
78. When 1 mol NaCl is dissolved in 1 kg water, expected ΔT_f (no dissociation) is 1.86 K but observed is 3.0 K. Approximate i is
- A. 1
 - B. 1.5–1.6
 - C. 2
 - D. 3
79. A 0.1 m solution of an unknown non-volatile nonelectrolyte has $\Delta T_f = 0.186$ K in benzene ($K_f = 5.12$). Approximate molar mass of solute if 1 g in 100 g benzene is
- A. $\sim 275 \text{ g mol}^{-1}$
 - B. $\sim 50 \text{ g mol}^{-1}$

C. $\sim 150 \text{ g mol}^{-1}$

D. $\sim 375 \text{ g mol}^{-1}$

80. If two solutions have same ΔT_f in same solvent and are both non-electrolyte, then

A. They have same number of moles of solute

B. They have same mass of solute

C. They have same molar mass

D. They have same density

81. (Reserve for additional specific JEE-style numeric if needed.)

SECTION 4: MIXED CONCEPTUAL (Q101–130)

101. An aqueous solution of urea is hypertonic to blood. Injecting it into bloodstream will cause RBCs to

A. Swell and burst

B. Shrink (crenate)

C. Remain unchanged

D. Become isotonic

102. An isotonic saline solution has same osmotic pressure as blood plasma. This is approximately

A. 0.9% NaCl (w/v)

B. 9% NaCl (w/v)

C. 0.09% NaCl (w/v)

D. 90% NaCl (w/v)

103. Reverse osmosis is used for

A. Softening hard water

B. Desalination of sea water

C. Increasing solubility of gas

D. Lowering osmotic pressure

104. In reverse osmosis, direction of solvent flow is

A. From solvent to solution

B. From more concentrated to less concentrated solution

C. From solution to pure solvent

D. No flow

105. If raw sea water has higher osmotic pressure than product water, to get reverse flow, applied pressure must be
- A. Less than osmotic pressure of sea water
 - B. Equal to osmotic pressure of sea water
 - C. Greater than osmotic pressure of sea water
 - D. Zero
106. Which of the following is NOT a colligative property?
- A. Osmotic pressure
 - B. Relative lowering of vapour pressure
 - C. Surface tension
 - D. Depression in freezing point
107. For two solutions A (0.1 m urea) and B (0.1 m NaCl, complete dissociation), which relation is correct?
- A. $\pi_A = \pi_B$
 - B. $\pi_A > \pi_B$
 - C. $\pi_A < \pi_B$
 - D. Depends on T
108. A solution boils at temperature lower than both pure components. It is likely
- A. Ideal
 - B. Positive deviation, minimum boiling azeotrope
 - C. Negative deviation, maximum boiling azeotrope
 - D. Non-volatile solute solution
109. A solution boils at temperature higher than both pure components. It is likely
- A. Ideal
 - B. Positive deviation
 - C. Negative deviation, maximum boiling azeotrope
 - D. Minimum boiling azeotrope
110. Which is correct?
- A. All azeotropes obey Raoult's law
 - B. Azeotropes have fixed boiling point and constant composition
 - C. Azeotropes can be easily separated by simple distillation
 - D. Azeotropes only form for solids

111. A solution of sucrose in water is
- A. Electrolyte solution
 - B. Non-electrolyte solution
 - C. Colloidal solution
 - D. Suspension
112. A 1 molal aqueous solution of any nonelectrolyte will have
- A. Same freezing point depression
 - B. Same boiling point elevation
 - C. Same osmotic pressure (if same T)
 - D. All of the above
113. For ideal solutions, activity of each component equals
- A. Its mole fraction
 - B. Its partial pressure
 - C. Its concentration in molality
 - D. Its molarity
114. For very dilute solutions, Raoult's law for solvent and Henry's law for solute together describe
- A. Ideal gas behaviour
 - B. Ideal solution behaviour
 - C. Ideal polymer behaviour
 - D. Only electrolytes
115. If two liquids are completely miscible, they form
- A. Always ideal solutions
 - B. Can be ideal or non-ideal
 - C. Always azeotropes
 - D. Always immiscible at high T
116. For liquid pairs showing strong specific interactions (like H-bonding) between unlike molecules, they likely show
- A. Positive deviation
 - B. Negative deviation
 - C. Ideal behaviour
 - D. Immiscibility

117. In ethanol–water solution, hydrogen bonding between unlike molecules is
- A. Weaker than like–like
 - B. Stronger than like–like
 - C. Same as like–like
 - D. Irrelevant
118. Solubility curve of most solids in water vs T is
- A. Increasing
 - B. Decreasing
 - C. Constant
 - D. Zig-zag
119. If a salt dissolves exothermically, increasing temperature
- A. Increases solubility
 - B. Decreases solubility
 - C. No effect
 - D. First increases then decreases
120. At saturation, rate of dissolution and rate of crystallization are
- A. Unequal
 - B. Equal
 - C. Zero
 - D. Undefined
121. In a supersaturated solution
- A. More solute is present than solubility limit
 - B. Less solute than solubility
 - C. No solute
 - D. Only solvent
122. Common ion effect generally
- A. Increases solubility
 - B. Decreases solubility
 - C. No effect
 - D. Random effect
123. For a sparingly soluble salt, adding a solution containing common ion will
- A. Precipitate more salt

- B. Dissolve more salt
 - C. No change
 - D. Make solution ideal
124. Henry's law breaks down when
- A. Gas reacts with solvent
 - B. Gas is very soluble
 - C. Pressure is high
 - D. All of these
125. In an ideal gas mixture, partial pressure is proportional to
- A. Mole fraction
 - B. Mass fraction
 - C. Volume fraction
 - D. Density
126. For a gas obeying Henry's law, doubling pressure
- A. Doubles mole fraction in solution
 - B. Halves mole fraction
 - C. Doubles K_H
 - D. Has no effect
127. For an ideal solution, activity coefficients
- A. Are 1
 - B. Are 0
 - C. >1
 - D. <1
128. In real solutions, activity of a component is product of
- A. Molarity and density
 - B. Mole fraction and activity coefficient
 - C. Molality and K_f
 - D. Vapor pressure and K_b
129. Which colligative property is most widely used in industry for purification (desalination, etc.)?
- A. Elevation in boiling point
 - B. Depression in freezing point

- C. Osmotic pressure (reverse osmosis)
- D. Relative lowering of vapour pressure

130. Total number of advanced MCQs in SOLUTIONS_LEVEL3 set is close to

- A. 80
- B. 100
- C. 130
- D. 200